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Enhancing sensor data reliability in structural health monitoring systems using digital twin technology

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Abstract

Structural health monitoring (SHM) systems are intended to collect continuous, current information on the condition of civic buildings, aircraft, and industrial gear. These structures may decay over time as a result of slow processes such as corrosion and fatigue, as well as unexpected catastrophes such as natural or man-made disasters. It is important to note, however, that the components of SHM systems may degrade after installation, resulting in incorrect information being supplied to decision-makers. To address this issue, digital twin (DT) technology has been utilized to provide an accurate representation of the building's actual state. Additionally, a weighted-Bayesian belief network (W-BBN) has been integrated into the framework to assess the building's condition. The framework also includes an example of maintenance and inspection throughout the lifespan of the structure, incorporating a sophisticated model of the SHM system to evaluate the level of risk. To show the proposed framework's operations, a numerical analysis and technique demonstration are offered, utilizing the building's smart sensors as a reference. The findings reveal that the unique SHM technology inspired by DT increases the identification of building health in an efficient and automated way.

KEYWORDS

artificial intelligence, digital twin, quality assurance, structure health monitoring

1 | INTRODUCTION

Deterioration of a structure is a dangerous phenomenon that can cause both chemical and physical harm. Environmental conditions, slow material degradation, and unexpected events can all play a role in the wear and tear of civil structures as time progresses. Maintaining structures requires a consistent strategy for routine inspections and maintenance.[1] A monitoring system is a sustainable approach that can minimize economic, health, and environmental hazards and extend the service life of a building.[2,3] Structural health monitoring (SHM) systems have been developed in recent years to provide information about the health of a building by identifying "damage-sensitive features." These features measure variations in damage indicators, which are categorized into three types: damage quantification, damage localization, and damage detection.[4] The degree of damage can be determined at multiple levels of detail based on the information and technology used for damage assessment. harm localization identifies a specific region, whereas quantification provides more exact information about the severity of the harm.

1.1 | Problem identification

The success of the SHM method for monitoring a building's health relies on the assumption that sensor data

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accurately reflects the structure's condition. However, it is important to consider the entire system, not just the structural components, when predicting performance.[5] Typically, a SHM approach includes a sensor module, data transmission module, and health evaluation module.[6] Unfortunately, harsh environmental conditions often cause sensors to malfunction, while internal system failures and wireless communication issues can result in data loss.[7] Such deviations can render the monitoring approach ineffective and lead to missing data in structural evaluation and management, resulting in significant financial losses. Additionally, anomalous information may lead to operational disruptions and structural repairs, increasing the likelihood of accidents.[5]

1.2 | State-of-the-art approaches

Various strategies are being developed to combat the deteriorating processes. The importance of sensor validation tools (SVTs) in assessing data dependability for chemical industry monitoring was initially realized by Dunia et al.[8] Later, Friswell and Inman[9] investigated the similar approach for SHM uses. Following that, researchers developed SVTs using single-class classification approaches and multivariate statistical methods.[5] One group of researchers investigates each sensor separately to determine if the monitoring system is functioning correctly,[10] while the second group analyzes the relationships between sensors in the network to identify sensor defects by comparing data from all sensors in the monitoring solution.[11–14] On the other side, there is an increasing interest in applying machine learning to uncover faults in diverse systems.[15] It has also been acknowledged that SVTs are prone to defects due to their limited operational capabilities and the difficulty in detecting certain irregularities in the data they collect. This poses risks that need to be addressed. However, considering that SVTs are typically expensive and require dedicated processing units, incorporating them into a SHM system needs to be evaluated in terms of its actual value. Additionally, in SHM applications, there is a challenge of convincing administrators and owners about the added value and societal benefits of implementing such systems.

Numerous research has focused on the value of information (VoI) derived from Bayesian decision theory to determine the possible long-term financial effect of a SHM system before its deployment.[16,17] This helps reduce management expenses resulting from collecting new knowledge.[18] Optimized methods of sensor network implementation,[19] maintenance,[20] and data collection strategies[21] are examples of these studies. Those interested can refer to a recent overview of the state of the art regarding VoI.[22] Information assessment is crucial for VoI evaluation as it is determined prior to the "real" SHM outcome.[23] Perfect and imperfect information are the two categories into which information is typically divided.[16] When complete information is available, the condition of a structure can be fully understood. However, when information is imperfect, uncertainty about the structure's condition remains. In the field of VoI, techniques such as cumulative error[24] and probability functions[25] are commonly used to model imperfect information. Simulation methods, both simple and complex, can be used to model imperfect information.[18,26] Some researchers have explored the relationship between data quality and VoI, but according to one study, VoI related to the condition of SHM systems has not yet been discussed.[27]

1.3 | Research gap

There are several categories of sensor malfunctions that can affect the accuracy of data due to various disrupting factors. In vibration-based SHM, sensor failures are classified into seven categories: drift, bias, low accuracy, gain, constant noise, baseline noise, and continuous recording.[28] The first four types are considered "soft" faults because they still provide some information about the structure, while the remaining three are classified as "hard" faults because they do not offer useful data. If these errors are not assessed for reliability before processing the collected data, they can cause severe malfunctions in monitoring systems. Current approaches for SHM primarily rely on pre-processed environmental factors rather than raw data, which can result in poor performance.[12] Therefore, it is crucial to develop an effective SHM strategy that addresses these limitations and facilitates preventive maintenance.

1.4 | Research contribution

Recent technological advancements, such as cloud infrastructure, 5G wireless networks, wireless sensor networks (WSNs), Internet of things (IoT) technology, machine learning (ML) algorithms, and high-performance computing, have helped to usher in an innovation based on information methodology known as the digital twin (DT). This technique has piqued the interest of academics from a variety of fields because of its capacity to build an accurate digital model of a physical item that evolves with it over time.[29] DT technology can analyze the condition of a structure or building by creating a virtual replica of the physical asset.[28,30] This digital duplicate may be

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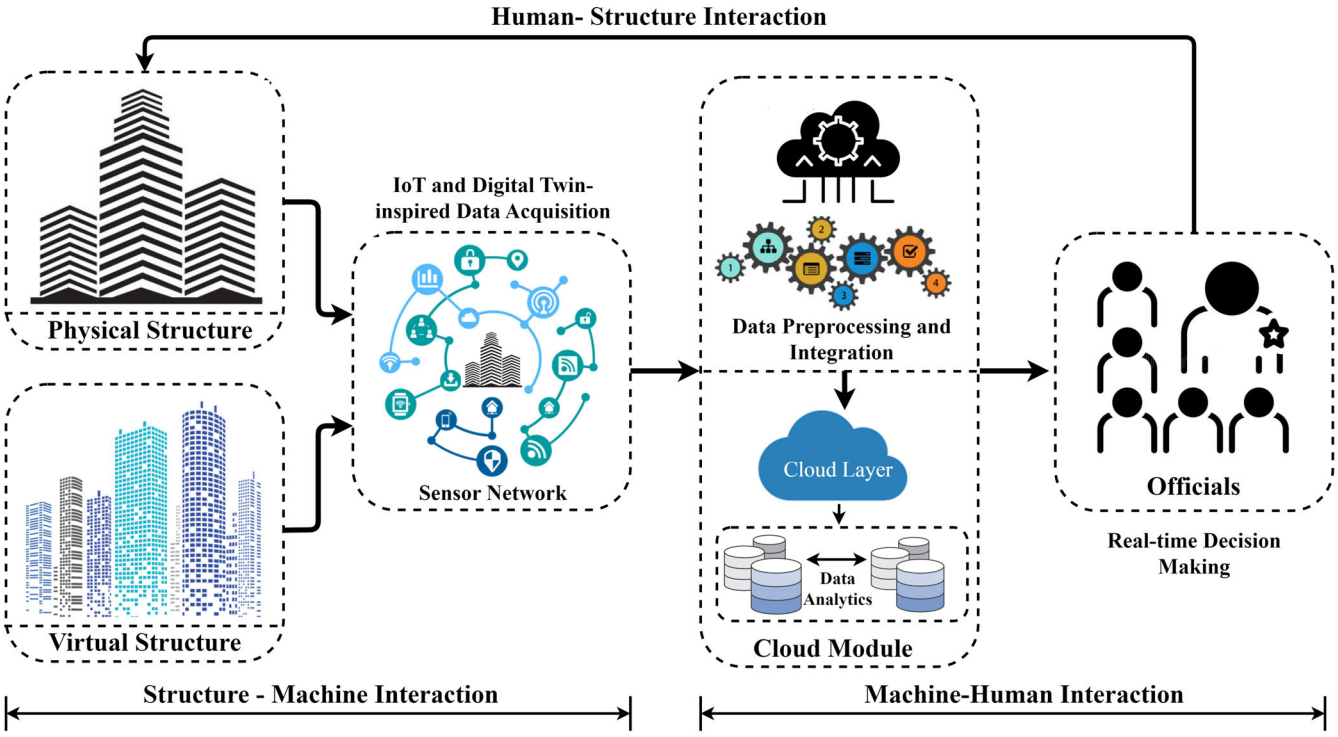


FIGURE 1 A conceptual approach of the proposed framework.

used to monitor performance, identify possible issues, and simulate various scenarios in order to enhance efficiency. The DT uses sensors and other sources of data to provide real-time information on the asset's state, allowing for predictive maintenance and early problem diagnosis. In this approach, the contribution of the proposed solution is shown in Figure 1 and can be summarized as follows:

1. Design and development of an autonomous IoT framework capable of monitoring the structural and environmental conditions.
2. Virtualization approach inspired by DTs is proposed to simulate the characteristics of structural and environmental parameters by analyzing several states of features.
3. Deployment of a probabilistic estimation solution based on weighted-naïve Bayes technique in cyberspace to assess the state of structures.
4. Structural condition assessment to achieve a qualitative assessment with respect to risk status.

1.5 | Article structure

Following Section 1, Section 2 showcases the theoretical and practical application of the framework by utilizing data from a real-life experiment. Furthermore, Section 3 includes multiple sensitivity analyses that assess different scenarios for implementation. Finally, Section 4 provides concluding remarks that emphasize the key findings.

2 | DIGITAL TWIN-INSPIRED STRUCTURE HEALTH DETERMINATION FRAMEWORK

The fundamental objective of this research path is to create a SHM system that precisely monitors the health of infrastructures in real-time while maintaining within a suitable budget for construction and operation.[31] This article introduces a framework called cloud-based digital twin for structure health determination (cDTSHD), developed through the collaborative efforts of the researchers belonging to the National Institute of Technology, Kurukshetra, and Lovely Professional University, Jalandhar. The cDTSHD platform is built using the Python programming language, chosen for its versatility and extensive support for scientific computing. It leverages powerful libraries such as TensorFlow for machine learning and pgmpy for probabilistic graphical modeling, facilitating robust data processing and analysis. The cDTSHD framework operates independently and is not based on existing DT platforms; rather, it has been developed from scratch to fulfill specific requirements for SHM. This design

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allows for customization and optimization tailored to the unique characteristics of various infrastructure types. The framework comprises several interconnected layers, including:

1. *Physical layer*: This layer interfaces with various sensors and IoT devices deployed on the infrastructure to gather real-time data. It employs protocols such as MQTT or HTTP for efficient data transmission and can handle both structured and unstructured data types.
2. *Virtual structure modeling layer*: Responsible for generating a virtual replica of the physical structure, this layer utilizes DT technology to create an accurate, dynamic model that reflects the real-time conditions of the infrastructure. By integrating data from the physical layer, allows for continuous updates and simulations, providing a comprehensive view of the structure's health and performance.
3. *Value of structure health (VoSH) quantification layer*: Utilizing machine learning algorithms, this layer processes the collected data to identify patterns and detect anomalies. TensorFlow is employed to train predictive models that assess structural integrity based on historical and real-time data inputs. Furthermore, visualization of data insights is achieved using libraries such as Matplotlib and Plotly, which provide interactive graphical representations of the health status of the structure. This layer enables stakeholders to visualize key performance indicators and trends over time, facilitating informed decision-making.

Moreover, automated operations within the cDTSHD framework are facilitated through task scheduling mechanisms, which allow for routine data collection and analysis without manual intervention. This automation enhances efficiency and ensures timely updates on the structural health status. However, some procedures still necessitate manual intervention, particularly during the initial setup and configuration of the layers. This hands-on monitoring guarantees that the system is properly modified to match the specific demands and climate of the buildings under observation. The model has been rigorously validated through a comprehensive case study, where both the DT and cloud platform were subjected to extensive testing. This validation demonstrated the capability of cDTSHD to deliver rapid and accurate computations, effectively supporting real-time SHM. Figure 2 illustrates the comprehensive process of assessing a structure's health, detailing the interactions between the various layers and the flow of data throughout the system.

2.1 | Physical layer

The physical layer of cDTSHD is divided into three connected subgroups: structural components, auxiliary elements, and measurement instruments. The primary structural components serve an important role in the framework's security and reliability. Furthermore, the client and the expert held bi-weekly conference calls to establish close interaction, where they discussed the format of the required measurements and the infrastructures available at each experimental site. As a result, the smart sensors were designed and developed to meet the building's requirements, providing a dependable, energy-efficient, and weather-resistant solution that addresses infrastructural gaps like electricity and connectivity. Therefore, the system requirements for analyzing a structure's health have been identified and listed below:

- measurements of temperature,
- measurements of humidity,
- temperature at the moisture point estimation,
- dry and moist conditions on the structure's walls.

After assessing the entire system's specifications, the intelligent structure of the system was created. Figure 3 illustrates the location where the smart tags have been installed throughout the designated study area. Moreover, the process of obtaining data is illustrated in Figure 4 and includes various components:

- *Main processing unit-ESP32*: A low-energy microcontroller that serves as the device's main computational unit and supports multiple peripheral communication interfaces. The device is equipped with the following features: a 32-bit LX6 microprocessor that can be single or dual-core and has a clock frequency of up to 240 MHz, 520 KB of SRAM, 448 KB of ROM, and 16 KB of RTC SRAM for memory, support for 802.11 b/g/n Wi-Fi connectivity with speeds of up to 150 Mbps for connectivity, and Ethernet, I2C, I2S, SPI, and UART for interfaces. It also has peripheral input/output options such as capacitive touch, analog-to-digital converter (ADC), digital-to-analog converter (DAC), controller area network (CAN) 2.0 inter-integrated circuit (I2C), universal asynchronous receiver/transmitter (UART), and serial peripheral interface (SPI).
- *Environmental and structural sensors*: (i) Samyoung SY-DS-1L is used to designate dry and moist states and communicate with the main processing unit via the ADC interface. The Samyoung SY-DS-1L is a sensor that measures humidity and can detect humidity levels ranging from 0% to 100% relative humidity (RH) with

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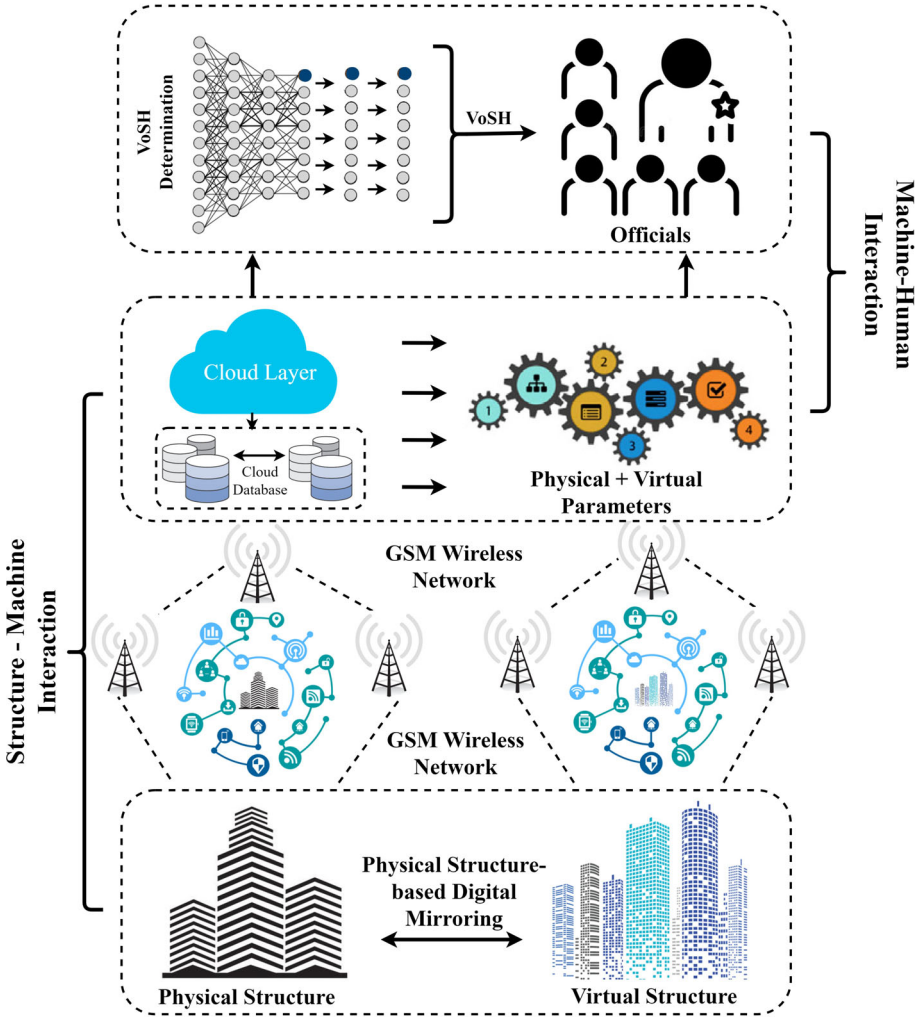


FIGURE 2 Workflow of the proposed framework.

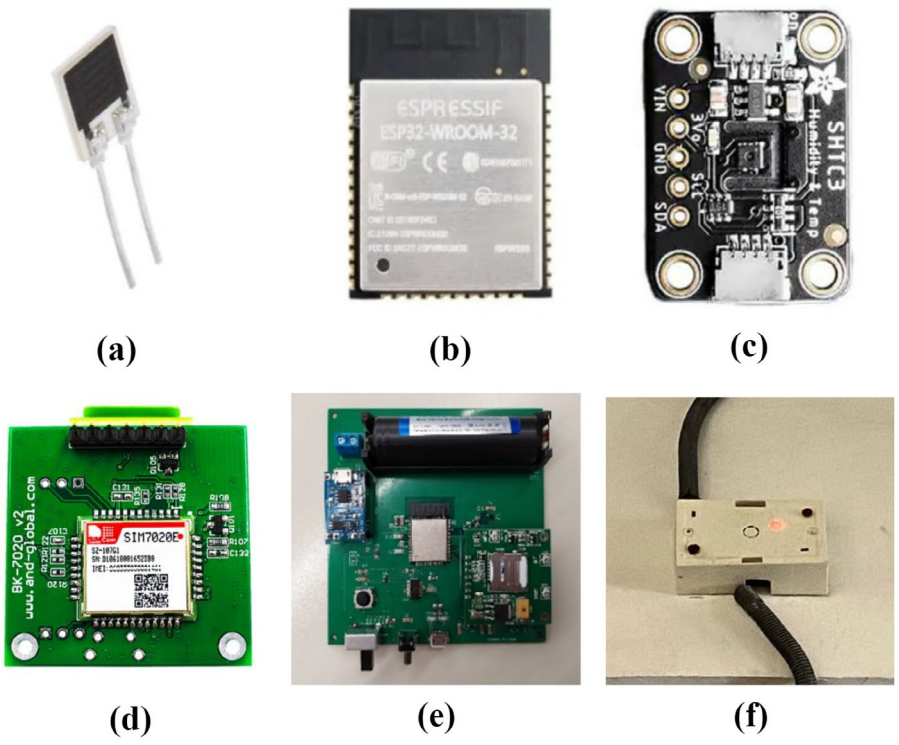


FIGURE 3 Smart tag components: (a) SAMYOUNG SY-DS-1L, (b) ESP32-WROOM32 SoC, (c) Sensirion SHTC3, (d) Simcom SIM7020E, (e) PCB assembly, and (f) smart tags deployment.

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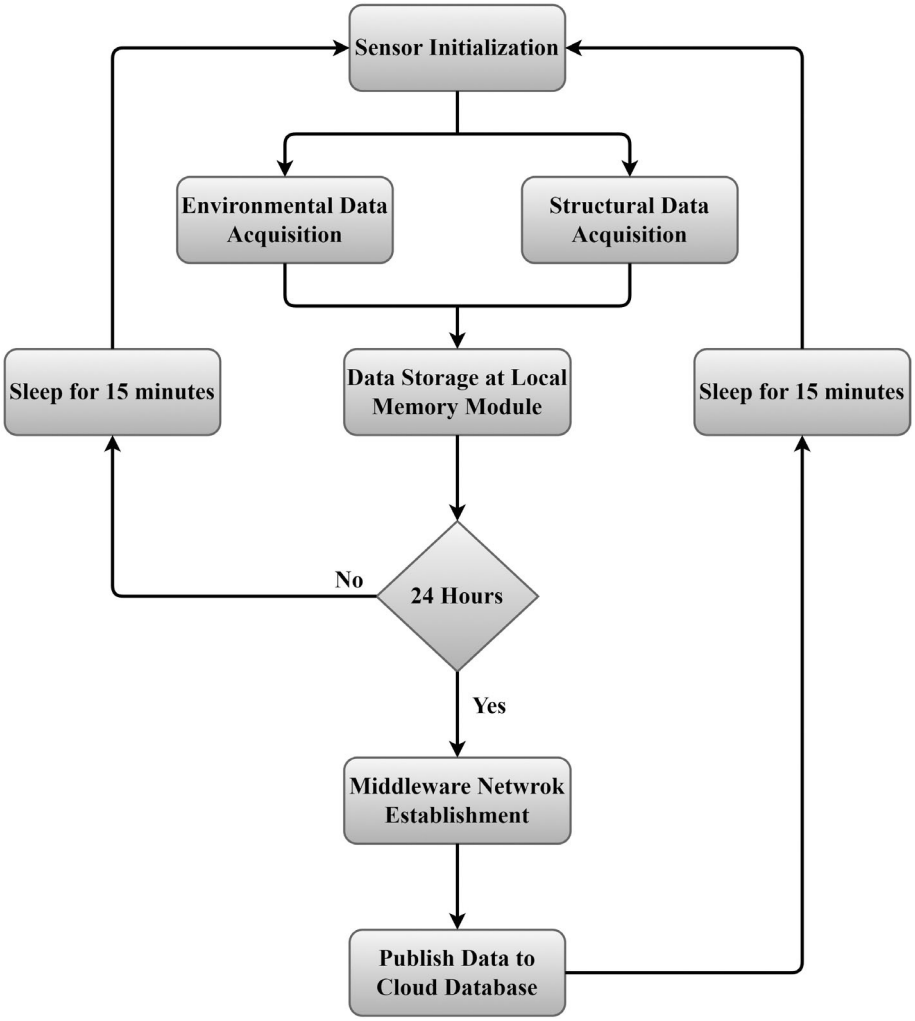


FIGURE 4 The working flow of the physical layer.

an accuracy of ±5% RH. It operates on a DC voltage of 800 mV and can function within a temperature range of 0–60°C. The sensor has a resistance of 10 kΩ and has a compact body with dimensions of 7 × 5.08 × 0.635 mm. (ii) Over an I2C interface, the Sensirion SHTC3 sensor measures temperature and relative humidity and communicates with the central processing unit. The device is capable of measuring humidity from 0% to 100% RH and temperature from −40 to 125°C. It has an accuracy of typically ±2% RH and ±0.2°C. The device powers on quickly and takes measurements in just 1 ms. It is traceable to NIST standards and has an I2C address of 0 × 70. The device is compatible with various supply voltages and has a low energy consumption of less than 1 μJ per measurement. It comes in a compact DFN package with dimensions of 2 × 2 × 0.75 mm^3. The device operates digitally within a voltage range of 1.62–3.6 V.

- *Power management unit*: Composed of a battery that can be recharged, a solar energy system, a recharging module, and a regulator of voltage that provides stable 3.3 V for the operation of the primary processing unit;
- *Network module*: The Simcom SIM7020E module provides connectivity through a 4G network and communicates with the main processing device via a universal asynchronous receiver/transmitter (UART) interface. The device has the following specifications: It can be controlled using AT commands such as V. 25TER, 3GPP TS 27.007, and SIMCOM AT Commands. The supply voltage range is between 2.1 and 3.6 V. It can operate in temperatures ranging from −40 to +85°C. The dimensions of the device are 17.6 × 15.7 × 2.3 mm, and it weighs approximately 1.3 g with a tolerance of ±0.2 g. The baud rate can range from 300 to 921,600 bps, with a default setting of 115,200 bps. The device has UART and GPIO interfaces. Firmware upgrades can be done through UART or FOTA. It is mounted using a B2B connector and has an SMT antenna interface.
- *Middle-ware*: The gateway collects data transmitted from the SIM7030 module over the IoT network using the message queuing telemetry transport (MQTT) protocol. In addition, it hosts the visualizing platform. It comes in a 40 PIN LCC package and has various

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hardware interfaces. Notably, it has the capability to enter PSM mode to minimize power consumption, which can extend battery life up to 10 years. Additionally, it should be noted that it comes in a 40 PIN LCC package.

2.2 | Virtual structure modeling layer

The proposed DT framework is compared to traditional architectures in Figure 5. It allows engineers to detect flaws in advance, improve the control system, and better prepare for preventing unexpected damage. Additionally, different infrastructure scenarios can be evaluated in a virtual environment before making actual changes. The framework consists of two phases: offline and online. The offline phase involves high-precision simulation, a knowledge library, a high-precision geometric model, and a data library. The online phase includes the diagnosis process, model modification, reusability assessment, and a data library. Each module is explained in more detail below:

1. *Geometric module*: This module includes the computer-aided design (CAD) model that was initially created and an updated geometric model that reflects the damaged features of the building for the high-precision simulation. The geometric condition of a building changes over time, and there may be differences between the final entity and the intended CAD model due to processing errors. Additionally, imperfections or damages will occur and need to be dynamically integrated into the geometric model.
2. *Simulation module*: The module includes simulation of multiple systems, simulation of multiple physics, modeling with reduced complexity, and modeling with surrogates. Since monitoring the health of a building is complex, its simulation model needs to consider multiple systems. A building is influenced by various environmental factors like air temperature, solar radiation, humidity, wind, precipitation, air pollution, and noise. To account for these factors, a multi-physics simulation can improve the accuracy of the model. It is also possible to use surrogate modeling techniques to speed up calculations without sacrificing precision. This module also serves as the model library for the knowledge library module.
3. *Knowledge library module*: This component can deliver a pre-defined degradation status to the diagnostic module and can be supplemented by the diagnosis module with the new degradation status.

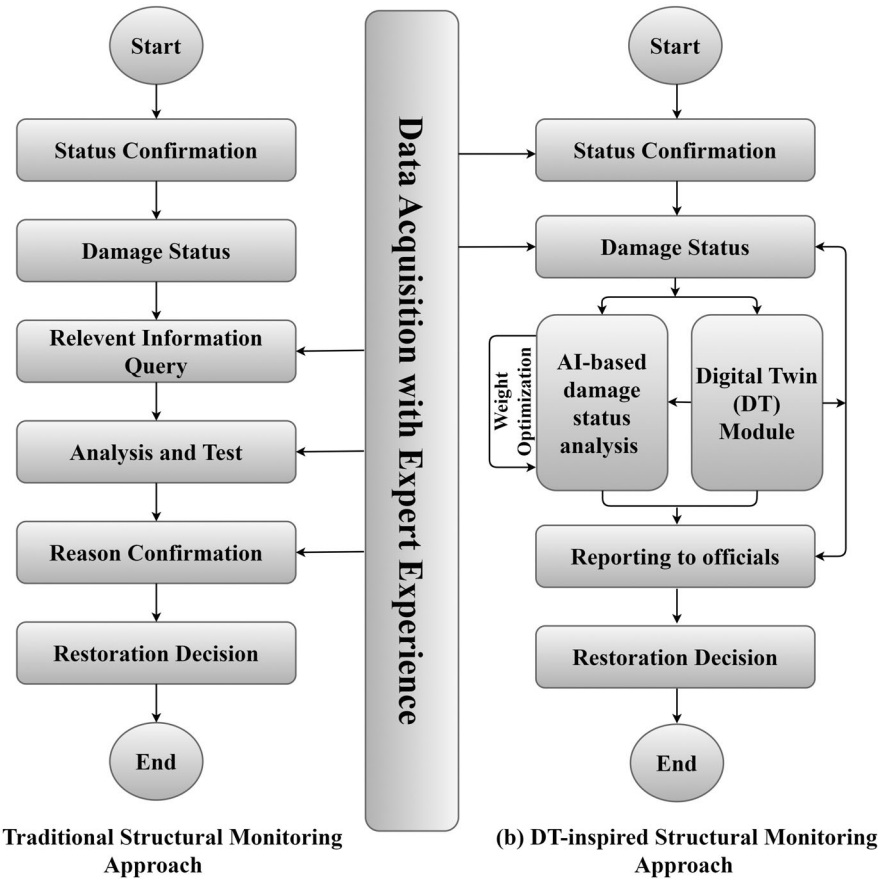


FIGURE 5 Difference between traditional and DT-inspired structural monitoring approach.

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This module also contains a variety of malfunction and failure modes as well as their corresponding methods of handling them.

4. *Offline data library module*: This component may interface with any other module in the offline stage to provide data interactions, which means that other modules can access and contribute to the data it stores at any time. It also saves information from all conceivable sources, including data sensing history and model-related data.
5. *Diagnosis module*: Supported by continuous monitoring data and advanced algorithms. This component is able to deliver details regarding degradation, as well as the various operations of data pre-processing, identification, detection, and evaluation, which are all part of the overall procedure of damage being diagnosed.
6. *Model updating module*: This module contains technologies for quantifying uncertainty and data-driven updating. Numerous uncertainties exist in the modeling, evaluation, and forecasting processes for the intended monument, including model hypotheses and mathematical algorithms, etc. Uncertainty quantification can be used to account for uncertainties from multiple sources.
7. *Evaluation module*: The evaluation module includes studies of deterioration, residual life, and dependability. Updated circumstances for the targeted variables, such as ambient temperature, moisture, wind, and the process of precipitation will improve the accuracy of the structural health deterioration assessment. The building's refurbishment can then be decided.
8. *Online data library module*: Information gathered from every source, including sensor monitoring data and state parameters in real-time. This particular module can be linked to any other online module and offer data interaction, similar to the offline data library module.

For instance, geometrical information is forwarded to the DT module to construct the model to perform simulation. Simulated data can then be used to identify failure modes, which can contribute to the development of a knowledge library. Additionally, relevant data can be traced back to identify errors in the simulation for diagnosis, model updates, and evaluating reusability. In summary, the benefits of utilizing DT technology in the proposed SHM framework can be outlined as follows:

1. The DT is distinguished by its functions (model update, diagnosis, data storage, and performance evaluation) and applications (structural health assessment, monitoring, prediction, and information sharing). The DT operations run in real-time, whereas the DT applications are started by the user and performed by combining several DT characteristics.
2. The framework has online and offline phases for model development, implementation, and maintenance. Physical and geometrical models are utilized to create low-order simulations for offline experimentation. Anomalies discovered during the online phase can be analyzed offline and directly related to the application's situation for SHM.
3. The DT model differs from typical simulations in that it may be modified in actual time, with a focus on updating the model of uncertainty. A practical framework has been developed for structure health management, which can serve as a foundation for future development.

2.3 | VoSH quantification layer

In this section, Figure 6 illustrates a general procedure for determining the VoSH for damage detection using physical and virtual data. The entire process is separated into two subtasks:

1. Event-based change analysis that attempts to identify the time instants at which the primary symptomatic parameters of successive operational information have been speculated to have changed as a result of operating condition variances or the emergence of anomalies;
2. VoSH determination attempting to distinguish the events resulting from variations via probabilistic event matching method.

The weighted-Bayesian belief network (W-BBN)[32] is a method for classifying data that can be used to determine the health of a building without predefined thresholds. It involves comparing changes in acquired events to identify those caused by targeted factors, eliminating false alarms caused by regular variations.[33] This technique can be used on any structure and allows for appropriate responses to be provided quickly. In this manner, VoSH determination no longer requires prior knowledge of operational procedures with the implementation of W-BBN methodology by using pgmpy library.[34] The W-BBN method calculates the chance $p(r_t|X)$ of the run length r_t using operational data $X = \{x_1, ..., x_t, ...\}$ from a given sensor. In this manner, the process refers to the number of observations since the last observed event with a change. If any change is noticed, the overall run length will be reduced to zero; otherwise, as a new change point X_t is observed, the value of the run length will be increased by one. Here, it is suggested that the last

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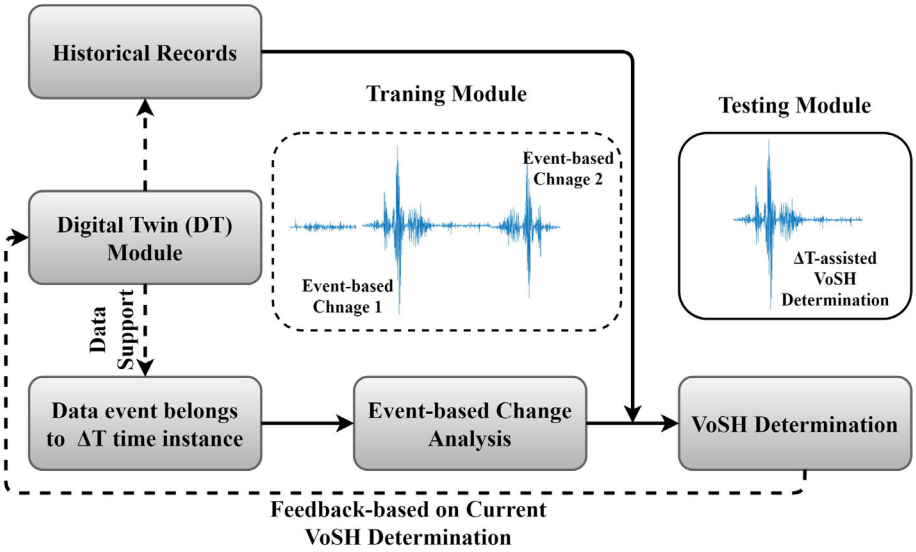


FIGURE 6 VoSH determination process.

change occurred at $t-r_t$ and the current run's observed data is represented as $X_t^r=\{x_{t-r_t+1},...,x_t\}$. Using Bayes' law,[35] the procedure for calculating the distribution that follows r_t can be expanded under the proposed solution:

$$\begin{aligned} p(r_t|X_{1:t}) &\propto p(X_{1:t},r_t)=\sum_{r_{t-1}} p(X_{1:t},r_{t-1},r_t) \\ &=\sum_{r_{t-1}} p(X_t,r_t|X_{1:t-1},r_{t-1})p(X_{1:t-1,r_{t-1}}) \\ &=\sum_{r_{t-1}} p(r_t|r_{t-1})p\left(X_t|X_t^r\right)p(r_{t-1},X_{1:t-1},) \end{aligned} \tag{1}$$

Here, $p(r_t|x_{1:t})$ is a function of $p(r_{t-1}|x_{1:t-1})$, indicating that the probability distribution can be calculated in a recursive manner. Furthermore, the scheme updates the value of distribution using two calculations: first, the probability related to the prior change point denoted as $p(r_t|r_{t-1})$ and the distribution of prediction $p\left(x_t|x_t^r\right)$ over the latest observation from the most recent data points.

To keep the process of probability calculation simple, it has been considered that the duration of each run follows the process of an exponential distribution. In this manner, the value of prior probability with respect to the change point is considered by the pre-defined value of hazard denoted by h which is independent of r_t and $p(r_t|r_{t-1})=h$. Moreover, the duration of run reset itself during $p(r_t|r_{t-1})=1-h$ when $r_t=r_{t+1}+1$. The value of probability distribution $p\left(x_t|x_t^r\right)$ is directly dependent on the process of generating x_t^r with respect to the most recent events with change. Particularly, the value of predictive distribution $p\left(x_t|x_t^r\right)$ is described by the finite number of statistics where generative distribution probability belongs to the conjugated exponential family. It has been assumed that x_t^r follows the process of Gaussian distribution with the unknown value of mean and variance represented as θ and λ, respectively. In this manner, the prior hyper-parameters $\eta_0=\{\mu_0,\kappa_0,\alpha_0,\beta_0\}$ The mean and variance values have been set to correspond to the general form of the Normal-Gamma distribution.

$$\begin{aligned} p(\theta,\lambda|\mu_0,\kappa_0,\alpha_0,\beta_0)=&\frac{\beta_0^{\alpha_0}\sqrt{\kappa_0}}{\Gamma(\alpha_0)\sqrt{2\pi}}\lambda^{\alpha_0-\frac{1}{2}} \\ &\exp\left(-\frac{\lambda}{2}\left[\kappa_0(\theta-\mu_0)^2+2\beta_0\right]\right) \end{aligned} \tag{2}$$

As new findings emerge gradually, the hyper-parameter set is modified in the following manner:

$$\begin{aligned} &\mu_t=\frac{\kappa_0\mu_0+tX_t^r}{\kappa_0+t},\ \kappa_t=\kappa_0+t,\ \alpha_t=\alpha_0+\frac{t}{2} \\ &\beta_t=\beta_0+\frac{1}{2}\sum_{i=t-r_t+1}^{t}\left(x_i-x_t^r\right)^2+\frac{\kappa_0 t\left(x_t^r-\mu_0\right)^2}{2(\kappa_0+t)} \end{aligned} \tag{3}$$

The posterior predictive distribution $p\left(x_t|x_t^r\right)$ is a generalized t-distribution with mean μ_t, variance $\frac{\beta_t(x+1)}{\alpha_t\kappa_t}$, and $2\alpha_t$ degree of freedom.

3 | CASE STUDY WITH EXPERIMENTS

The proposed approach is used to study the effect of different environmental parameters on a building's health. The decision-making process involves choosing between a healthy state (C_1) and an unhealthy state (C_2), and between taking no action (A_1) or requiring renovation

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TABLE 1 Utility determination.

Action	C_1—healthy	C_2—not healthy
A_1—do nothing	0	1
A_2—renovation	−1	0

(A_2). According to Table 1, the combination of actions and states related to the structure's condition is associated with a utility. If the structure is working and in good health, it is considered to have no loss, resulting in a utility value of zero. However, if the structure is functional but damaged, it requires renovation, and the utility value is defined as 1. Various experiments were conducted and discussed in this manner:

1. Detail of civil structure
2. Data acquisition efficiency
3. System stability analysis
   - Network strength analysis
   - Power consumption analysis
4. VoSH determination with cost analysis
   - VoSH determination with performance validation
   - Time factor-based cost estimation
   - Rate of error (RoE) with parametric correlation analysis
5. Comparative analysis

3.1 | Detail of civil structure

The suggested architecture was initially evaluated at Lovely Professional University in India, namely at the Department of Computer Science and Applications (DCSA) building. This three-story structure has a total size of 30,500 square feet and comprises classroom, office, research, and testing spaces. The created DCSA building DT has five layers, combines several data sources, and aids in damage detection. The present case study demonstrates how the suggested technology may help with the data integration process for a dynamic DT of existing structures, assist its condition analysis function, and investigate further potential and problems.

3.2 | Data acquisition efficiency

To evaluate the reliability of the smart tags, humidity and temperature readings obtained from the tags are compared against data sourced from a commercial weather station. Specifically, historical environmental data were acquired from a commercial online platform,* which provides a weather API that aggregates meteorological data from a network of weather stations distributed globally. The temperature and humidity measurements captured by the Smart Tags and the commercial weather stations are illustrated in Figure 7a,b, respectively. Additionally, the DT framework leverages sophisticated simulation methodologies that merge real-time data with predictive modeling techniques. By employing open-source libraries such as SimPy[36] for discrete-event simulation and OpenFOAM[37] for computational fluid dynamics, the framework is capable of simulating a diverse array of scenarios. These libraries facilitate the model's ability to incorporate various operational conditions, environmental factors, and potential failure modes, thereby enabling a comprehensive assessment of the system across numerous conceivable events. Furthermore, the acquired data is analyzed using the pandas[38,39] libraries, which provide robust tools for data manipulation and numerical computation.

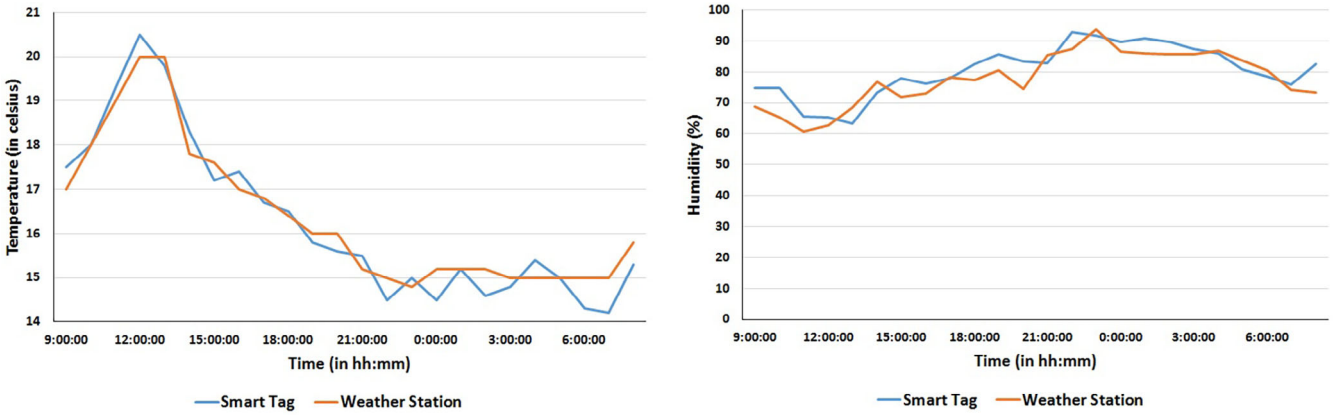
The results show that the data collected by the IoT device is remarkably similar to that collected by the meteorological station, with minimal differences during the research day. In fact, the biggest and lowest temperature fluctuations between the smart tag and the weather station are roughly 1.7 and 0.1°C. At certain times of the day, the weather station displays the highest variation of 10% and the lowest variation of 0.2% for humidity data. These deviations can be partially attributed to the geographical distance between these devices and the lack of decimal precision in the data provided by the weather stations.

3.3 | System stability analysis

Connectivity and resource efficiency are two of the fundamental features and system requirements of the proposed framework. Regarding connectivity, the primary limitation of the sites is the absence of WiFi networks at the majority of the historic sites. In addition, it was evident that a low-power device-based network in an IoT network is necessary because the embedded sensor needed to function for an extended period of time. Regarding energy efficiency, many traditional monuments and buildings lack access to a power source; thus, other sources of energy need to be utilized. The following tests were carried out and reported to establish the best solutions for the above-mentioned challenges.

*Source: https://www.visualcrossing.com (accessed on January 25, 2022).

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(a) Temperature acquisition analysis

(b) Humidity acquisition analysis

FIGURE 7 Data acquisition analysis.

3.3.1 | Network strength analysis

The narrow band (NB) IoT network was selected as the main network preference because it is particularly designed for IoT applications, making it small, low-power, and more efficient at low data transmission rates. In comparison to long-term evolution (LTE), this network may link devices with a received signal strength indicator (RSSI) of −105 dB. The signal strength is monitored to determine the connectivity range of the IoT SIM7020 module used in the smart tag. The connection strength is assessed in a three-story structure with a terrace and a basement at Lovely Professional University in India. Measurements were collected every 30 s for 5 min on the terrace, second level, first level, ground floor, and basement. Ten RSSI measurements were taken on each floor and plotted in Figure 8. It was discovered that on the terrace, where the gadget is not blocked by walls, the signal intensity is stronger, reaching −85 dB. However, there is no substantial change in RSSI values between the first and second levels. The signal weakens on the bottom floor and reaches −110 dB in the basement, where it is at its weakest. Despite the poor signal, the gadget can talk with the IoT network and send the data it has gathered. This illustrates that the IoT network can communicate over great distances and often function in locations where 4G/3G connectivity is limited.

3.3.2 | Power consumption analysis

To attain the aim of power efficiency and maintain the sensor network's extended operational life, particular hardware design options have to be explored. The smart sensors' equipment includes a battery that can be charged, a solar energy source, an electrical tracking module, and a USB interface. As a result, it can be powered by various sources of energy and can operate indefinitely on solar energy. Every power mode utilized by the device is enumerated in order below.

- *Normal mode*: The system initializes the instruments, gathers, and saves data.
- *Sleep mode*: To save power, the bulk of the device's components are turned off.
- *Data transmission*: The instrument establishes a network connection and transmits the stored data.

A digital multimeter from Pro's Kit MT-1705 was used to measure the consumption of power by the smart tag in order to estimate its power profile. The measurements were recorded every second. The smart tag had been configured to collect and transfer environment measurements every minute for this test. For the time frame of 1 min, it assumes a situation of inactivity. The device's 2-min power consumption graph is depicted in Figure 9. In normal mode, the device's average current consumption is approximately 36 mA, while in the state of inactivity, it is 0.06 mA. In this manner, the most demanding process is data transmission, which reaches an extreme level of 86 mA.

3.4 | VoSH determination with cost analysis

With the assistance of a smart sensors-based network, an extended period of data is incorporated into the DT module, making it possible to a preliminary assessment of the health condition of the building. To test to evaluate the relative efficacy of the suggested approach, a data set with a 1-h sample period is used. The determination of VoSH and performance analysis is done as follows.

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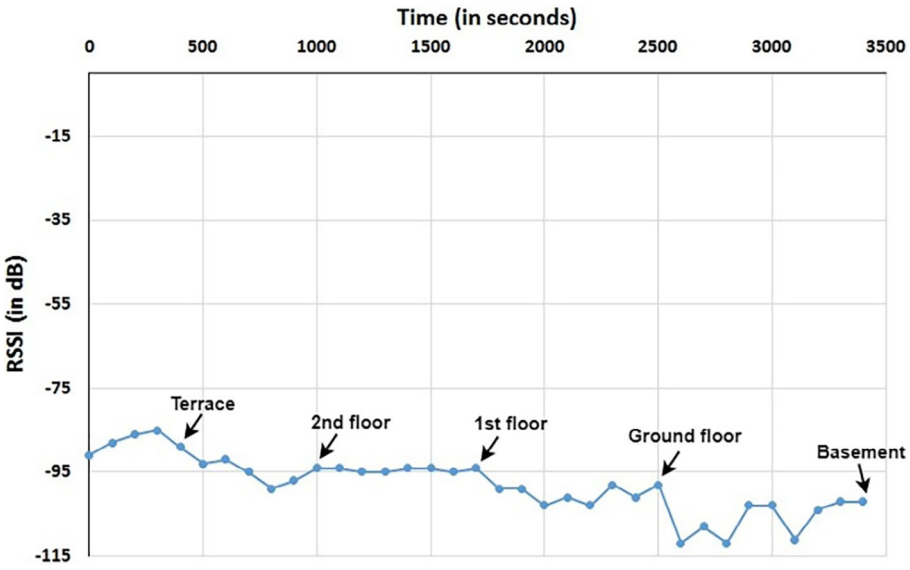


FIGURE 8 Network strength analysis.

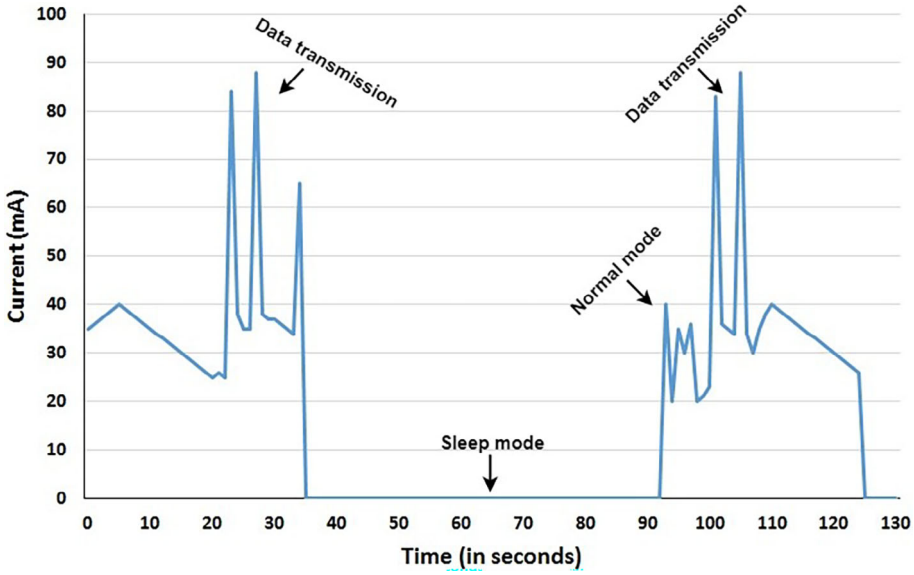


FIGURE 9 Power consumption analysis.

3.4.1 | VoSH determination with performance validation

The suggested approach for VoSH determination is evaluated using a variety of performance criteria, including precision, specificity, sensitivity, and F-measure. These measures are critical because they enable a comprehensive evaluation of model performance, especially in instances when class imbalance may exist. Accuracy measures the framework's overall soundness, whereas sensitivity assesses its capacity to reliably recognize positive cases. Specificity complements sensitivity by assessing the model's ability to recognize negative cases. The *F*-measure combines accuracy and recall into a single statistic, offering a balance that is especially useful in situations where false positives and false negatives have distinct consequences. Furthermore, the data used to train the model comprises both normal and aberrant events related to VoSH. This dataset is critical for verifying that the model can effectively generalize to new data, particularly in anomaly detection. To successfully evaluate the effectiveness of the model, the dataset is split into training and test sets, usually in an 80:20 ratio. This fraction provides enough training while maintaining a large amount of data for impartial testing. The training set is used to fit the model, while the test set evaluates its predicted performance. In addition, three baseline classifiers are used to assess the accuracy of VoSH determination: random forest tree (RFT), decision tree (DT), and *k*-nearest neighbor (k-NN) using the scikit-learn module.[40] These classifiers are selected due to their diverse algorithmic approaches, which allow for a robust comparison.

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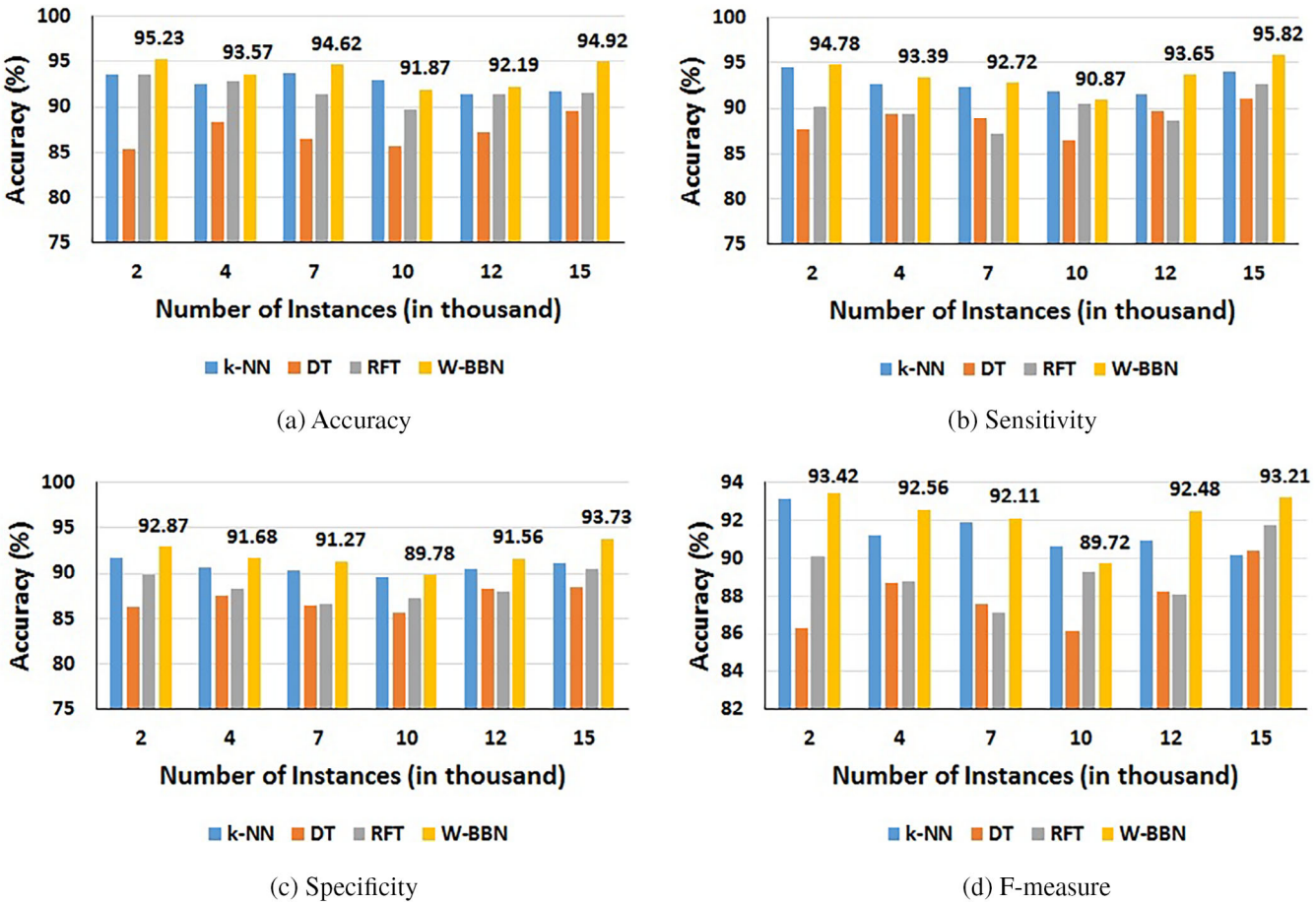
Decision Trees provide a clear visualization of decision-making processes, while Random Forests enhance accuracy through ensemble learning. *k*-NN offers a simple yet effective method for classification based on proximity, making it a valuable benchmark.[41,42]

Furthermore, genuine and false positive and negative instances are evaluated based on the predictions made by the model vs. the actual labeling in the test set. For this purpose, the dataset includes examples of both normal behavior and various anomalies, enabling the model to learn the distinguishing features of each class. This comprehensive evaluation ensures that the model not only performs well on standard cases but is also capable of detecting anomalies effectively. When comparing the efficacy of the suggested solution to state-of-the-art methodologies, it is vital to note that only the parameter categorization solution was altered during the comparison analysis. This focused approach ensures that the impact of the proposed solution can be isolated and evaluated effectively. In contrast, all other parameters remained in their original settings, thereby maintaining consistency and fairness in the comparison. This methodology underscores the reliability of the results and highlights the significance of the proposed solution in Figure 10.

The W-BBN filter is implemented at the cloud layer to divide the gathered parameters into two categories: healthy and unhealthy, Figure 10 depicts the efficacy of the W-BBN classifier. Its calculated outcomes clearly define the VoSH determination efficacy of the proposed W-BBN classifier over the selected state-of-the-art classifiers. The W-BBN classifier has achieved an accuracy of 94.92%, as compared to the *k*-NN (91.72%), DT (89.54%), and RFT (91.48%). Regarding the Specificity measure, the proposed solution recognizes true positive events by attaining an accuracy of 93.73%, which is considerably higher than the accuracy of existing state-of-the-art classifiers. A similar enhancement is also evident in sensitivity and *F*-measure, with accuracies of 95.72% and 93.21%, respectively, both of which significantly surpass the success rates of the state-of-the-art classifiers.

3.4.2 | Time factor-based cost estimation

The Python-based Ubidots framework is employed to categorize the targeted events and generate alerts in accordance. Figure 11 depicts the time taken by each classifier to categories occurrences into the specified class. The



(a) Accuracy

(b) Sensitivity

(c) Specificity

(d) F-measure

FIGURE 10 Performance validation by performing comparative analysis.

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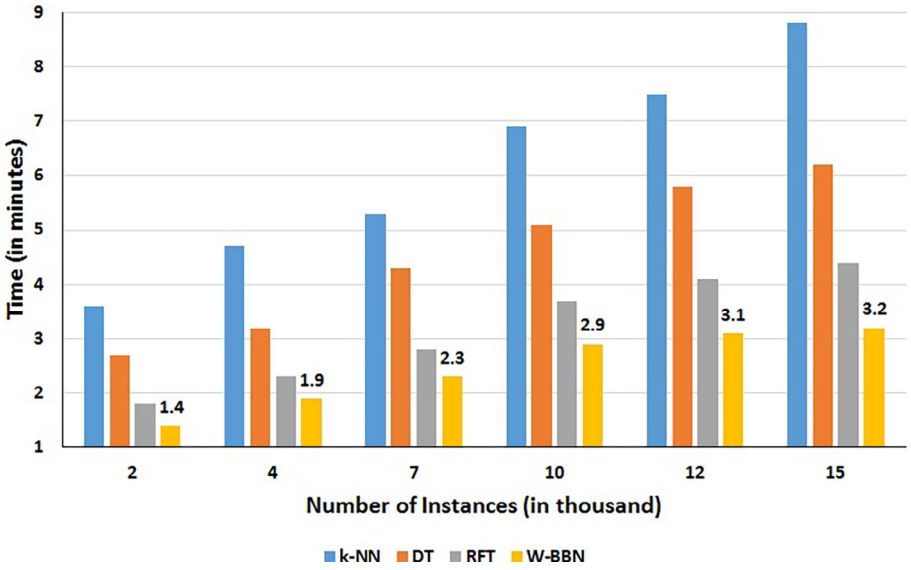


FIGURE 11 Classification time analysis.

readings show that the suggested classifier takes less time than existing state-of-the-art classifiers, emphasizing the responsiveness of the proposed approach. Furthermore, it can be shown that the proposed technique is better suited to handling massive amounts of data. As a result, the suggested system is substantially more successful at making fast decisions during monitoring in order to assist limit harm.

3.4.3 | Rate of error with parametric correlation analysis

By doing a comparative study, it can be determined that including a new classifier in the suggested framework is significantly easier and helps to satisfy the needs of the application. However, to analyze the event determination stability of the proposed model, different error and correlation-based performance matrices such as are calculated and exhibited in Table 2. Based on the projected error rates and correlation values, it is reasonable to conclude that the suggested W-BBN classifier is the best tackle for the current implementation case.

3.5 | Comparative analysis

Table 3 compares the existing frameworks with the proposed solution. It is clear that industrial solutions offer a wide range of functionalities, albeit at a high cost. The existing frameworks primarily utilize physics-based models, which can only analyze the condition of the building in an offline mode. Additionally, some data-driven models can detect the presence of damage. In contrast, the proposed model not only identifies damage but also provides crucial information regarding the specific location of the structural issues. It is important to note that quantitative assessments remain challenging due to the inaccessibility of the frameworks and the relevant data. The table highlights two key reasons why the development and operational costs of the proposed framework are categorized as low to moderate:

1. The DT layer contributes to the framework's stability by simulating every conceivable event to determine the next possibility.
2. The data analytics module is built on open-source and free libraries.

In addition, the proposed solution is capable of performing multiple SHM operations with an increasing degree of complexity. By following the approach of a DT, the proposed solution can combine real and simulated data that helps to detect the existence of damage with its location.

4 | DISCUSSION AND LIMITATION

A smart structure health monitoring solution is urgently needed to quickly analyze the health of a building and take preventative measures before severe consequences occur. While efforts have been made to automate the analysis of a structure's health, it is mainly done manually by experienced professionals. However, advanced analytical tools, such as machine learning or artificial intelligence, can effectively distinguish different patterns. The challenge lies in the reliance on IoT sensors, which can generate anomaly data due to environmental factors. To overcome this limitation, a framework inspired by DTs is proposed for structure health monitoring. This

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TABLE 2 Performance analysis parameters.

Performance parameters	*k*-NN	DT	RFT	W-BBN
RMSE (%)	9.31	11.42	14.54	8.56
MAE (%)	8.52	9.15	13.65	7.18
Correlation coefficient	0.9365	0.8572	0.8343	0.9798
RAE (%)	10.36	17.78	20.41	9.84
Average consumed time (each fold)	86.62	27.67	13.93	13.21

TABLE 3 State-of-the-art comparative analysis.

Parameters	Shim et al.[11]	Qian et al.[13]	Jeong et al.[14]	Tran-Ngoc et al.[12]	Proposed framework
Data type	Various	Time-series data	Time-series data	Tabular data	Time-series data
Data collection	Manual	Semi-automatic	Semi-automatic	Manual	Automatic
Data storage	Local server	Cloud database	Cloud database	Local server	Cloud database
Data analytic	Data-driven model	Data-driven model	Data-driven model	Physical factor analysis	Data-driven model
Digital twin	No	No	No	No	Yes
Portability	Local machine	Multi-devices	Multi-devices	Local machine	Multi-devices
Cost	Not applicable	Not applicable	Medium to high	Low	Low to medium
Monitoring	Periodic	Periodic	Near real-time	Periodic	Near real-time

involves mounting various sensors on the building's walls to monitor its condition. The use of DT technology helps manage and analyze data collected from different sources efficiently. Although the proposed solution can monitor the health of outdoor structures, it is important to also monitor the indoor environment and conditions. For example, changes in room temperature during winter can also affect the health of the structure. Therefore, indoor structure condition monitoring should also be developed within the DT framework for a more comprehensive evaluation of structural health.

5 | CONCLUSION

Monitoring frameworks for the health of structures often provide modified data for decision-makers, but there is a lack of published works on modeling information quality for structure health analysis. This research aims to address this gap by proposing a structure monitoring framework inspired by DT technology and incorporating the Weighted-Naïve Bayes theory to enable effective analysis of a structure's health. The framework improves on existing models by accounting for irregular sensor conditions, such as noise and missing information, through the use of virtual events. In cases where there are no SHM information available, virtual events inspired by DTs are generated to address faulty sensor conditions. In order to demonstrate the usefulness of the framework, a case study was conducted and various performance measures were calculated. The key findings of the proposed solution are as follows: (1) the SHM framework coupled with the DT generally provides greater benefits compared to traditional SHM systems; (2) the use of a DT eliminates the impact of noise on the SHM results; (3) the suggested solution's results are greatly influenced by prior knowledge about the structure's condition and SHM systems, as reflected in the previous chances, as well as value of utility associated with various combinations of activities and conditions of the architecture. In this sense, our work contributes to the current body of information by creating an innovative approach that allows researchers to systematically monitor a building's health using DTs. Future studies will look at further practical uses of DTs to support a wider range of services.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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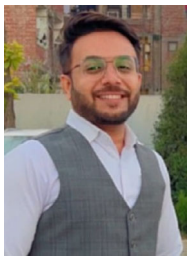
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